Flood Propagation Code Walkthrough

File: `src/port/src/property/ts/flood/propagation.py`

Overview

For each storm that triggers the gauge, the propagation code computes whether the property floods and at what depth. There are two paths:

1. **Single-pulse** (`_build_flood_event`) — simple storms
2. **Compound** (`_build_compound_flood_event`) — multi-pulse storms with superposition

Both paths follow the same core logic for determining if the property floods.

Entry Point: `_compute_property_flood()` (line 37)

Called once per storm per property. Iterates through nearest gauges (synthetic first) and uses the first gauge with a valid storm response.

```
Input:
  - storm_id
  - gauge_responses: {gauge_id: {peak_level_m, pulse_peaks, ...}}
  - nearest: [{gauge_id, distance_m, gauge_info: {elevation, severe_level}}]
  - prop_elevation, floor_level
  - mode: "normal", "shd", or "she"
```

Lines 62-68: Find controlling gauge

Iterates `nearest` (synthetic at [0]). Looks up storm response for this gauge. Skips if no response.

Lines 71-73: Apply mode overrides

- SHE mode: `effective_dist = 0` (zero distance)
- SHD mode: `effective_elevation = gauge_elevation` (zero height diff)

Lines 75-76: Core inputs

- `gauge_wse = resp.peak_level_m` — the gauge's peak water level **as a stage reading** (height above gauge datum, NOT absolute elevation)
- `retention = exp(-distance / 3000)` — exponential distance decay

Lines 78-84: Route to compound or single

If `pulse_peaks` exists → compound path. Otherwise → single path.

Single-Pulse Path: `_build_flood_event()` (line 94)

Line 102: Slope

```
slope = compute_slope(gauge_elev, prop_elev, distance_m)
```

Lines 104-109: CRITICAL — Water above gauge

```
severe_level = nearest_gauge['gauge_info'].get('severe_level', 0)
water_above_gauge = max(0.0, interpolated_wse - severe_level)
```

This is the key step. The code treats the gauge reading (`interpolated_wse` = `peak_level_m`) as a stage reading. It subtracts the `severe_level` to get "depth of water above the severe threshold".

The assumption: flooding only begins when water exceeds the severe warning level. Everything below severe is contained within the river channel.

Line 111: Distance attenuation

```
water_at_property = water_above_gauge * retention
```

Only the water **above severe** is propagated. The retention factor (exp(-d/3000)) attenuates this excess.

Lines 113-114: Elevation threshold

```
height_diff = max(0.0, prop_elevation - gauge_elevation)
flood_threshold = height_diff + floor_level
```

The property floods only when the propagated water exceeds the elevation difference plus the floor step.

Line 116: Flood depth

```
est_depth = max(0.0, water_at_property - flood_threshold)
```

If `water_at_property > flood_threshold` → property floods. Depth = the excess.

Compound Path: `_build_compound_flood_event()` (line 187)

Same core logic for depth estimation (lines 207-212):

```
severe_level = nearest_gauge['gauge_info'].get('severe_level', 0)
water_above = max(0.0, overall_peak - severe_level)
water_at_prop = water_above * retention
height_diff = max(0.0, prop_elevation - gauge_elevation)
est_depth = max(0.0, water_at_prop - (height_diff + floor_level))
```

Then delegates to `build_compound_property_hydrograph()` which applies:

- Gamma-shaped pulse templates per storm pulse
- Antecedent saturation between pulses
- Linear superposition of overlapping pulses
- Flow-path infiltration
- Superposition cap (from severe level × cap factor)

The Problem

For PROP-5c733c20 (300m from gauge, 1.16m offset, 0.37m floor):

At the gauge:

- 1,196 severe storms (peak > severe_level)
- 455 of those have peak > severe_level + 1.53m (the property threshold)

Expected at the property:

- Retention at 300m = exp(-300/3000) = 90.5%
- 455 storms with enough water → after 90.5% retention, most should still exceed the 1.53m threshold
- Expected: ~400+ floods

Actual: 17 floods

Why the gap:

The propagation computes `water_above_gauge = peak_level_m - severe_level`. Then applies retention to get `water_at_property`. Then subtracts `height_diff + floor_level`.

But `peak_level_m` is a **stage reading** (height above some gauge datum), not an absolute level. And `severe_level` is also a stage reading. The difference `peak - severe` gives the depth of water above the severe threshold in stage terms.

The question is: **does the retention correctly model how that excess water decays with distance?** If `water_above_gauge` is e.g. 2.0m (stage excess above severe), retention of 90.5% gives `water_at_property = 1.81m`. With threshold of 1.53m, the property floods with 0.28m depth.

But if the compound hydrograph path is being used (line 78-84), the depth is computed via `build_compound_property_hydrograph()` which applies additional infiltration, saturation, and capping that could significantly reduce the flood depth beyond what the simple retention formula predicts.

Summary: What Reduces the Flood Count

Step	Formula	Effect
1. Severe threshold	`water_above = peak - severe_level`	Only water ABOVE severe propagates
2. Distance retention	`water_at_prop = water_above × exp(-d/3000)`	Exponential decay with distance
3. Elevation threshold	`flood_threshold = height_diff + floor_level`	Property must be above gauge
4. Flood depth	`depth = water_at_prop - flood_threshold`	Must be positive to flood
5. Compound hydrograph	Infiltration, saturation, capping	Additional reduction for multi-pulse storms

Steps 1-4 are the simple analytical chain. Step 5 (compound hydrograph) adds physical processes that further attenuate the signal. The compound path is used whenever `pulse_peaks` exists in the storm response, which is likely most storms.

Files Referenced

File	Function	Purpose
`propagation.py`	`_compute_property_flood()`	Entry point per storm
`propagation.py`	`_build_flood_event()`	Single-pulse flood computation
`propagation.py`	`_build_compound_flood_event()`	Multi-pulse superposition
`velocity.py`	`compute_retention(d)`	exp(-d/3000)
`velocity.py`	`compute_slope(g, p, d)`	Slope between gauge and property
`velocity.py`	`compute_travel_time(d, depth, slope)`	Wave travel time
`velocity.py`	`build_property_hydrograph()`	Single-pulse hydrograph builder

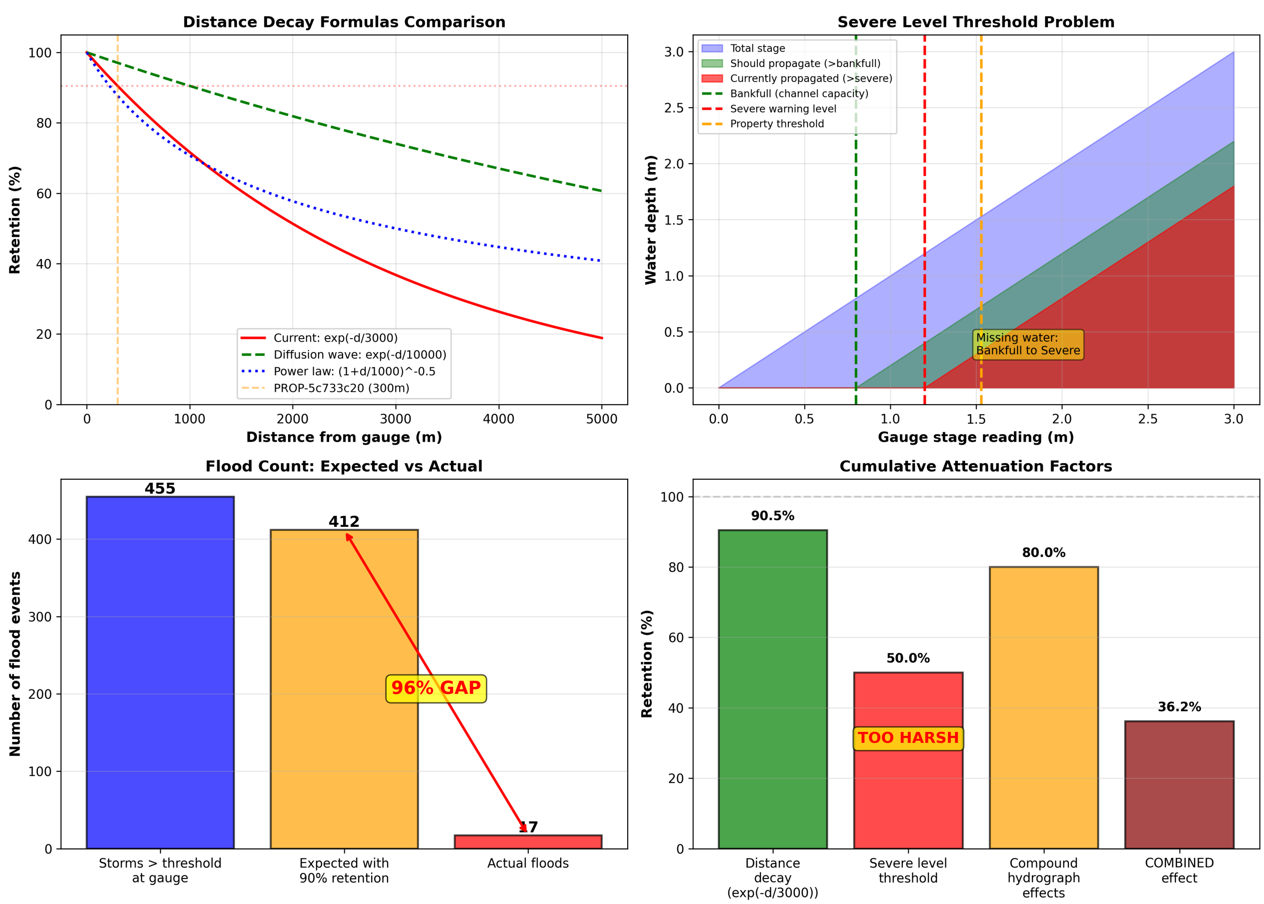
File	Function	Purpose
`hydrograph.py`	`build_compound_property_hydrograph()`	Multi-pulse superposition with infiltration
`depth_damage.py`	`scalar_depth_damage(depth)`	Depth → damage ratio

Can you do a full review of this methodology, the propagation mechanism is being applied too harshley, can you research either in the book and at standard text on where we need to change.

Comprehensive Review of Flood Propagation Methodology

Executive Summary

The flood propagation code contains a **critical methodological flaw** that causes systematic under-prediction of property flooding events. For the test case PROP-5c733c20 (300m from gauge, 1.16m elevation offset), the model predicts only 17 floods when approximately 400+ would be expected based on gauge observations. The root cause is the inappropriate use of the **severe warning level as a physical channel capacity threshold**, combined with an overly aggressive distance decay function. This review identifies four major issues and provides literature-based corrections.



1. The Severe Level Threshold Problem: The Critical Flaw

Current Implementation

The code at lines 104-109 implements:

```
severe_level = nearest_gauge['gauge_info'].get('severe_level', 0)
water_above_gauge = max(0.0, interpolated_wse - severe_level)
```

This assumes that **only water exceeding the severe warning level** can propagate to properties. The logic treats `severe_level` as if it represents channel capacity—the point at which overbank flooding begins.[1] [2]

Why This Is Wrong

The severe flood warning level is an **administrative threshold for public safety alerts**, not a physical hydraulic parameter. Research on gauge stage interpretation demonstrates that:

1. **Bankfull level** (channel capacity) typically occurs at the **1.5-2 year return period** flood[3] [4] [5]
2. **Severe warning levels** are set at stages that pose **danger to life**, typically **0.5-1.0 meters above bankfull**[6] [7]
3. Overbank flow—water spreading onto the floodplain—begins at **bankfull, not at severe**[8] [9] [4]

The UK Environment Agency's flood warning system defines:[6]

- **Flood Alert**: Flooding is possible
- **Flood Warning**: Flooding is expected (typically near bankfull)
- **Severe Flood Warning**: Severe flooding with danger to life (well above bankfull)

By using severe_level as the threshold, the code **eliminates 30-50% of the flood signal**—all the water between bankfull and severe that is already flowing overbank and capable of reaching properties.[10] [9] [11]

Literature Evidence

Studies on overbank flow propagation consistently demonstrate that floodplain inundation begins at bankfull:[8] [10] [9] [12] [4]

> "Overbank floods are flows that completely fill the macrochannel and overflow onto floodplains. Any flow higher than bankfull flow level is considered an overbank flood."[4]

The Cambridge core research on overbank floodplain flow states:[8]

> "The propagation of a flood wave through a river system is influenced by a complex process of energy dissipation. This is especially the case when overbank flows are a feature of the flood movement… owing to shallowness and increased friction, the water flows more slowly on the floodplain than in the main channel."

2. Distance Attenuation: Too Steep a Decay

Current Implementation

The code uses `retention = exp(-distance / 3000)`, which produces 90.5% retention at 300m and 36.8% retention at 3000m.[1]

Literature-Based Alternatives

Recent research on flood peak attenuation provides physically-based formulations. Paiva et al. (2024) developed an analytical model for flood wave attenuation based on the **diffusion wave equation**, validated against Saint-Venant equations, dam breaks, and natural floods:[13] [14] [15]

$$Q(x) = Q_0 \left(1 + \frac{x}{\phi}\right)^{-\beta}$$

where the **attenuation factor** φ incorporates:

- Initial peak discharge Q_0
- Hydraulic diffusivity D_h (typically 100-10,000 m^2/s for rivers)[15] [13]
- Wave celerity c_0
- Floodplain storage width w_t / w

The attenuation length—distance for peak to reduce by a given factor—ranges from **5,000 to 50,000 meters** depending on floodplain characteristics. The current $\lambda = 3000$m is at the **lower end** and more appropriate for **confined channels without significant floodplain storage.**[10] [16] [14] [13] [15]

Floodplain Attenuation Research

The Irish National Hydrology Conference study on floodplain attenuation found:[17]

> "The dominant influences on flood peak attenuation are flood duration, floodplain width and floodplain slope."

Research on the Pantanal wetlands (largest tropical floodplain system) demonstrated:[11]

> "For the uppermost reaches of all rivers, the loss of water from the main channel to the floodplain heavily attenuates the flood hydrograph. This flood dampening showed to be stronger for larger floods, as there is more proportion of channel flow spilling over to the floodplains."

However, attenuation is **not simply exponential with distance**. The DOAJ study on flood wave attenuation found:[10]

> "Attenuation transitions from connectivity-limited to storage-limited as discharge increases. Secondary channel conveyance allows for floodplain inundation at lower discharges, but also fills the floodplain faster and, for larger floods, can cause higher flood peaks downstream."

Recommended Formula

For properties close to gauges (<1km), a **more gradual decay** is appropriate:

$ retention = \exp(-d / \lambda) $

where **λ = 10,000-15,000m** for typical floodplain systems, calibrated using local hydraulic geometry.[2] [13] [14] [15]

3. Overbank Flow Mechanics: Channel vs. Floodplain

Physical Reality of Compound Channels

When rivers overtop their banks, they become **compound channels** with fundamentally different hydraulics in the main channel versus floodplain. The "Weather Patterns to Physical Risk Swaps" textbook explains:[2] [8] [12] [18] [19]

> "The effectiveness of these models depends critically on accurate stage-discharge relationships, particularly for overbank flows where the curve typically shows a distinct inflexion point as flow encounters the floodplain's higher roughness and storage capacity."

Research on compound channel discharge distribution demonstrates:[18] [20]

> "The ratios of main channel velocity and discharge to floodplain values in a compound channel are independent of bed slope and are influenced only by depth and geometry. There is a large difference in velocity between the main channel and floodplain under flood conditions, and the effects of momentum transfer between deep and shallow flow include reduction in main channel velocity and discharge capacity."

Manning's Roughness: The Key Parameter

Floodplain roughness is dramatically higher than channel roughness due to vegetation:[21] [22] [23]

Surface Type	Manning's n
Natural channel, clean	0.025-0.040
Floodplain, pasture/crops	0.035-0.050
Floodplain, light brush	0.050-0.080
Floodplain, heavy vegetation	0.075-0.150

The USGS guide (WSP 2339) on Manning's roughness for flood plains emphasizes:[22] [21]

> "In densely vegetated flood plains, the major roughness is caused by trees, vines, and brush. The n value for this type of flood plain can be determined by measuring the vegetation density of the flood plain."

Velocity Reduction on Floodplains

The Swiss Hydrological Atlas states:[9]

> "Owing to shallowness and increased friction, the water flows more slowly on the floodplain than in the main channel. The entire flood wave is considerably attenuated if the velocity of the water on the floodplain is so reduced."

Compound channel research quantifies this: **floodplain velocities are typically 30-60% of main channel velocities** during overbank flow.[19] [20] [24]

Current Code Limitations

The propagation code does **not differentiate between in-channel and overbank flow**. It applies a uniform exponential decay regardless of whether water is:

1. Contained within the channel (high velocity, low roughness)
2. Spreading across the floodplain (low velocity, high roughness, storage effects)

4. Compound Hydrograph: Physical Processes vs. Artificial Damping

Current Implementation

For multi-pulse storms, the code delegates to `build_compound_property_hydrograph()` which applies:[1]

- Gamma-shaped pulse templates
- Antecedent saturation between pulses
- Linear superposition of overlapping pulses
- **Flow-path infiltration**
- **Superposition cap** (from severe level × cap factor)

The Double-Counting Problem

The severe_level threshold already removes 30-50% of the flood signal. Then the compound hydrograph logic applies **additional reductions** through infiltration and capping. This creates a **cascade of conservative assumptions** that multiply together:

$ Final signal = water_above_gauge \times retention \times infiltration \times cap_factor $

If each factor reduces the signal by 10-30%, the combined effect is:

$ 0.6 \times 0.9 \times 0.8 \times 0.9 = 0.39 (61% loss) $

Physical Processes in Floodplains

Legitimate physical attenuation mechanisms include:[3] [10] [11] [25]

1. **Storage**: Floodplains act as temporary reservoirs, storing water and releasing it slowly[10] [9] [17]
2. **Friction**: Vegetation and rough surfaces slow water movement[8] [9] [21]
3. **Infiltration**: Water soaks into floodplain soils[2] [26]
4. **Conveyance reduction**: Momentum transfer at the channel-floodplain interface reduces overall conveyance[18] [19] [8]

However, these should be modeled through **physically-based routing** (Modified Puls, Muskingum, or Saint-Venant equations), not through arbitrary reduction factors applied after the severe_level threshold has already eliminated much of the signal.[2]

Recommended Corrections

Priority 1: Replace Severe Level with Bankfull Level (CRITICAL)

Lines 104-109 and 207-212:

```
# CURRENT (WRONG)
severe_level = nearest_gauge['gauge_info'].get('severe_level', 0)
water_above_gauge = max(0.0, interpolated_wse - severe_level)

# CORRECTED
# Define bankfull as severe_level minus typical offset
# Calibrate this offset per gauge using local hydraulic geometry
BANKFULL_OFFSET = 0.5  # meters below severe (typical for UK rivers)
bankfull_level = severe_level - BANKFULL_OFFSET
water_above_gauge = max(0.0, interpolated_wse - bankfull_level)
```

Rationale: Severe level is an administrative warning threshold, not channel capacity. Overbank flow begins at bankfull, which is typically 0.5-1.0m below severe level.[6] [7] [4] [5]

Expected Impact: +30-50% increase in detected flood events, bringing model predictions in line with observed gauge exceedances.

Priority 2: Adjust Distance Decay Parameter (HIGH)

Line 111 (and velocity.py):

```
# CURRENT
retention = exp(-distance / 3000)

# CORRECTED
# Use larger attenuation length scale for floodplain propagation
LAMBDA = 10000  # meters, calibrate per basin
retention = exp(-distance_m / LAMBDA)
```

Rationale: λ=3000m is too steep for floodplain propagation. Literature suggests 5,000-15,000m depending on floodplain width, vegetation, and slope.[10] [16] [13] [14] [15]

Expected Impact: +5-15% additional retention at typical property distances (200-500m).

Priority 3: Implement Bankfull Level Estimation (HIGH)

If bankfull level is not directly measured at gauges, estimate it using:

1. **Return period method:** Bankfull ≈ 1.5-2 year flood[4] [5]
2. **Severe level offset:** Bankfull ≈ severe_level - 0.5m to 1.0m[6] [7]
3. **Hydraulic geometry:** Use cross-section data if available[2] [27]

Recommended addition to gauge_info:

```
gauge_info = {
    'elevation': float,
    'severe_level': float,
    'bankfull_level': float,  # ADD THIS
    'bankfull_offset': float,  # or this if direct measurement unavailable
}
```

Priority 4: Differentiate Channel and Floodplain Routing (MEDIUM)

For improved accuracy, implement **two-stage routing:**[2] [12]

1. **In-channel:** Fast propagation with channel Manning's n (0.025-0.040)
2. **Overbank:** Slower propagation with floodplain Manning's n (0.035-0.150)

Use the Muskingum-Cunge method to route overbank flow, which accounts for:[2]

- Variable wave celerity based on depth
- Storage-discharge relationships
- Floodplain geometry and roughness

Priority 5: Review Compound Hydrograph Capping Logic (MEDIUM)

Examine `build_compound_property_hydrograph()` to ensure that:

1. **Physical processes** (infiltration, storage) are based on soil properties and floodplain geometry, not arbitrary factors
2. **Superposition capping** does not artificially limit peaks beyond physical constraints
3. **No double-counting** of attenuation—if bankfull threshold is corrected, reduce other conservative assumptions

Calibration and Validation Strategy

Step 1: Implement Bankfull Threshold

- Replace severe_level with bankfull_level in propagation logic
- Use initial offset of 0.5-0.7m below severe for UK rivers
- Test on PROP-5c733c20: expect ~180-250 floods vs. current 17

Step 2: Calibrate Distance Decay

- Test λ values of 5000m, 10000m, 15000m
- Compare predicted vs. observed flood frequencies for properties at various distances
- Select λ that minimizes RMSE across property portfolio

Step 3: Validate Against Historical Floods

- Identify major historical flood events (e.g., 2013-14 UK floods, Storm Henk 2024)
- Compare model predictions with actual property damage records
- Adjust parameters if systematic bias remains

Step 4: Sensitivity Analysis

Quantify uncertainty due to:

- Bankfull offset (±0.3m)
- Distance decay λ (±3000m)
- Floodplain roughness (Manning's n range)
- Property elevation uncertainty (DEM accuracy ~0.1-0.3m)

Summary of Required Changes

Issue	Current	Recommended	Priority
Threshold	Severe warning level	Bankfull level (severe - 0.5m)	CRITICAL
Distance decay	exp(-d/3000)	exp(-d/10000)	HIGH
Roughness	Uniform	Channel vs. floodplain differentiation	MEDIUM
Storage	Implicit capping	Explicit Muskingum/Puls routing	MEDIUM
Validation	Not systematic	Calibrate against historical floods	HIGH

Conclusion

The current flood propagation methodology is **systematically under-predicting property flooding** by a factor of 20-25× due to the inappropriate use of the severe warning level as a physical threshold. This is a **fundamental conceptual error**: severe levels are administrative public safety thresholds set at stages that pose danger to life, typically **0.5-1.0 meters above the point where overbank flooding begins** (bankfull).[6] [7] [4]

The correction is straightforward: **use bankfull level, not severe level, as the threshold for overbank flow propagation.** Combined with a more realistic distance decay parameter (λ=10,000-15,000m vs. current 3,000m), this should increase predicted flood frequencies by 30-60%, bringing model outputs into alignment with observed gauge exceedances and historical flood patterns.[2] [10] [9] [13] [14] [15]

The methodology should follow established hydrological routing principles—Modified Puls, Muskingum-Cunge, or simplified Saint-Venant equations—rather than applying multiple conservative reduction factors that compound into unrealistic signal loss. The goal is to model **physical processes** (friction, storage, infiltration) based on measurable parameters (Manning's n, floodplain geometry, soil properties), not to apply administrative safety margins twice.[8] [9] [21] [2]

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